

Software Architecture Layout

1 Memo Summary

Purpose of document: define the software architecture and execution flow used to complete line tracking, rescue, drop-off, and homing.

Purpose of calculation in this architecture: the fuzzy scheduling calculation determines mode-dependent speed limits and PID gains from sensor-derived inputs X_1 and X_2 .

Result of calculation used by architecture: each loop computes a mode label and parameter set $(v_{\max}, K_p, K_i, K_d)$ that directly drives motor command generation.

How this supports team design decision: the state-machine structure separates mission phases, and the adaptive control calculation is reused in each line-following phase (LINE_READING, GOTO_ZONE, HOMING) to keep behavior consistent and robust.

2 Software Plan

Using camera + IR/color sensor combination.

Infrastructure

- Use a state machine to track the current program stage.
- Start in LINE_READING, which fuses IR array signals and the camera frame to compute motion commands.
- If only one line is detected: calculate the line length L_k .
- Else: calculate the intersection first, then compute the line length L_k .
- Use L_k and the current speed to obtain the fuzzy mode and PID parameters.

3 Design Specifications (Software)

Item	Specification	Purpose
Control architecture	Finite state machine	Mission sequencing and fault isolation
Sensor fusion	IR array + camera ROI	Combine lateral error with lookahead visibility
Fuzzy inputs	X_1 and X_2 in 0–100	Normalized scheduler inputs
Controller output	$(v_{\max}, K_p, K_i, K_d)$	Adaptive speed cap and PID gains
Motor command bounds	left/right in $[-1,1]$	Safe command saturation
Line-following reuse	LINE_READING, GOTO_ZONE, HOMING	Consistent behavior across phases

4 Calculation Link to Architecture

At every control loop where line following is active:

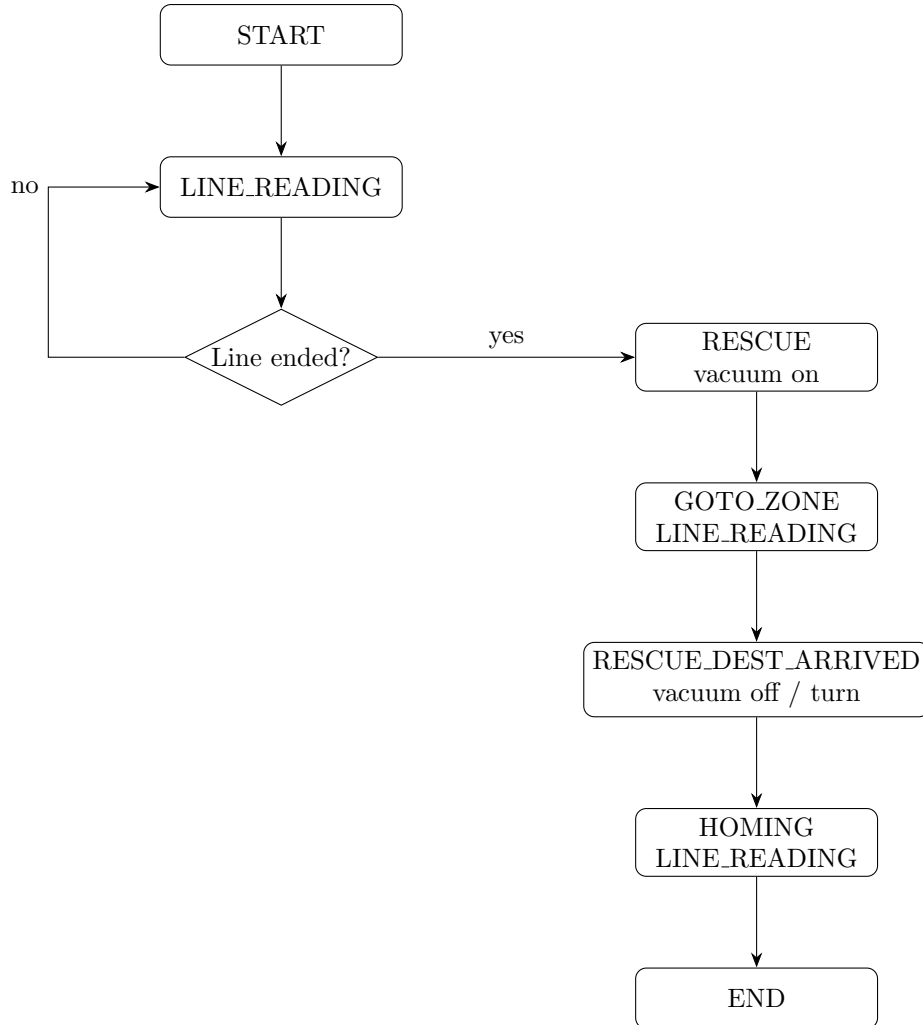
$$X_1 = \frac{\max_j x_{j,k}}{|X|} \times 100, \quad X_2 = \frac{L_k}{|L|} \times 100$$

The fuzzy block returns the mode-dependent tuple $(v_{\max}, K_p, K_i, K_d)$, then motor commands are computed by:

$$u = K_p e + K_i \int e dt + K_d \frac{de}{dt}, \quad u_L = \text{clamp}(v - u), \quad u_R = \text{clamp}(v + u)$$

This ties the design calculation to state-machine execution instead of treating the algorithm as an isolated module.

5 State Machine Flowchart



6 Node-by-Node Behavior

- **START:** initialize sensors, camera, communication, and motor driver; set initial mode and variables.

- **LINE.READING**: run the closed-loop line follower by fusing IR error and camera line-length information to generate left/right motor commands.
- **Line ended?**: decision node; if the main line is no longer detected, transition to rescue sequence, otherwise continue **LINE.READING**.
- **RESCUE**: stop or slow as required, activate the vacuum mechanism, and secure the Lego figure.
- **GOTO_ZONE / LINE.READING**: navigate to the drop-off zone using the same line-reading controller (IR + camera + fuzzy/PID scheduling).
- **RESCUE_DEST.ARRIVED**: perform destination action (optional turn based on zone orientation), then deactivate vacuum to release the Lego figure.
- **HOMING / LINE.READING**: follow the return path home using line reading again, with mission-complete constraints if needed (e.g., reduced speed).
- **END**: stop motors, place actuators in safe state, and terminate or idle.

7 LINE_READING Loop (Camera + IR Fusion)

```

loop:
  ir = read_IR_array()
  err = line_position(ir) - center
  pid_u = PID(err, Kp, Ki, Kd)

  frame = camera_capture()
  lines = detect_line_vectors(frame)
  Lk = compute_primary_line_length_or_intersection(lines)

  X1 = speed_percent(current_speed)
  X2 = lookahead_percent(Lk)

  mode = fuzzy_or_table(X1, X2)
  baseSpeed, (Kp, Ki, Kd) = mode_parameters(mode)

  left = clamp(baseSpeed - pid_u)
  right = clamp(baseSpeed + pid_u)
  set_motors(left, right)

```

8 Notes

- X_1 is the current speed command normalized to 0–100.
- X_2 is the visible line length percentage computed from the camera ROI.
- The state machine transitions to **RESCUE** when the line ends, then to **GOTO_ZONE**, **RESCUE_DEST.ARRIVED**, and **HOMING**.